SPIRTLAR

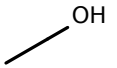
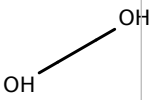
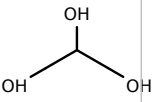
Gidroksil guruhli birikmalar · R–OH

Student book – nazariy qo‘llanma

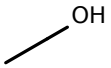
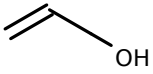
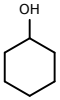
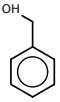
Uglevodorod radikali bilan **gidroksil guruhi (–OH)** birikishidan hosil bo‘lgan organik moddalar **spirtlar** deyiladi. Gidroksil – spirtlarning **funksional guruhi**. Gidroksil guruhlari soniga qarab spirtlar **bir atomli**, **ikki atomli** va **uch atomli** (hamda ko‘p atomli) bo‘ladi.

1. Tasnifi

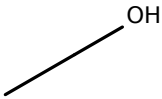
A) Gidroksil guruhlari soni bo‘yicha

Turi	Umumiy formula	Misol	Skelet
Bir atomli	$C_nH_{2n+1}OH$	etanol C_2H_5OH	
Ikki atomli (glikollar)	$C_nH_{2n}(OH)_2$	etilenglikol $C_2H_4(OH)_2$	
Uch atomli	$C_nH_{2n-1}(OH)_3$	glitserin $C_3H_5(OH)_3$	
Ko‘p atomli	—	sorbit, ksilit (4–6 ta –OH)	

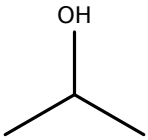
B) Uglevodorod radikali bo‘yicha

Turi	Misol	Skelet
To‘yingan	C_2H_5OH – etanol	
To‘yinmagan	$CH_2=CH–CH_2OH$ – allil spirt	
Siklik	$C_6H_{11}OH$ – siklogeksanol	
Aromatik	$C_6H_5–CH_2OH$ – benzil spirt	

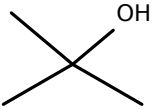
C) –OH birikkan uglerod bo‘yicha



birlamchi: etanol



ikkilamchi: propanol-2



uchlamchi: 2-metilpropanol-2

Birlamchi spirtda –OH **CH_2** guruhida ($R–CH_2–OH$), ikkilamchida **CH** guruhida ($R_2CH–OH$), uchlamchida **C** atomida ($R_3C–OH$) bo‘ladi. Metanol ham birlamchi hisoblanadi.

2. Bir atomli to‘yingan spirtlar

Umumiy formulasi: **$C_nH_{2n+1}OH$** yoki **$C_nH_{2n+2}O$** ($n \geq 1$); $M_r = 14n + 18$.

2.1 Tuzilishi

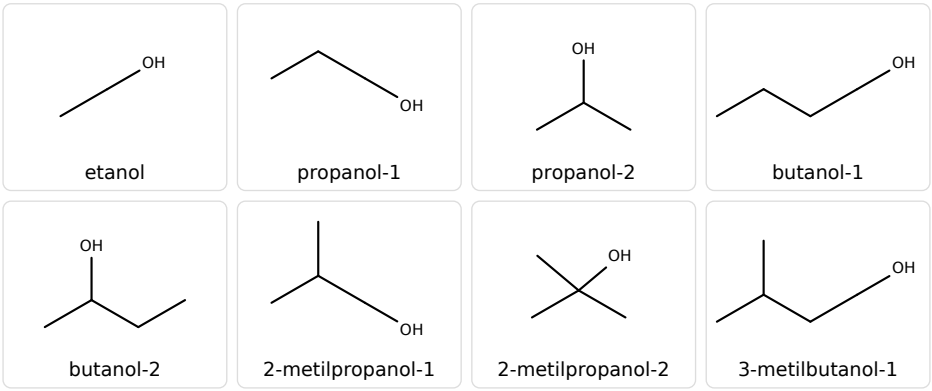
- Uglerod va kislorod atomlari **sp^3-gibridlangan**; C–O–H burchagi ≈ 109°.
- O–H va C–O bog‘lari **kuchli qutbli**: kislorod elektromanfiyroq, vodorod esa spirtda qisman musbat zaryadlanadi.
- Molekulalar orasida **vodorod bog‘** hosil bo‘ladi (O–H···O), shuning uchun spirtlarning qaynash harorati yuqori.
- σ-bog‘lar soni 3n + 2 ta, π-bog‘ yo‘q. Jami atomlar 3n + 3 ta.

2.2 Gomologik qator va nomlanishi

Formula	Sistematik nomi	Tarixiy / ratsional nomi
CH_3OH	metanol	metil spirti, yog‘och spirti
C_2H_5OH	etanol	etil spirti, vino spirti
C_3H_7OH	propanol-1	propil spirti
C_4H_9OH	butanol-1	butil spirti
$C_5H_{11}OH$	pentanol-1	amil spirti
$C_6H_{13}OH$	geksanol-1	geksil spirti

Sistematik nomenklatura (IUPAC)

- Gidroksil guruhi bor **eng uzun zanjir** tanlanadi; nomi uglevodorod nomiga **«-ol»** qo‘shimchasi qo‘shib yoziladi.
- Zanjir **–OH ga eng yaqin** uchidan raqamlanadi (OH radikaldan muhimroq); –OH ning o‘rni oxirida raqam bilan ko‘rsatiladi (propanol-2).
- Radikallar alifbo tartibida, –OH raqamidan oldin yoziladi (2-metilbutanol-2).
- **Ratsional nomi:** radikal nomi + «spirt» (etil spirt, izopropil spirt, tert-butil spirt).

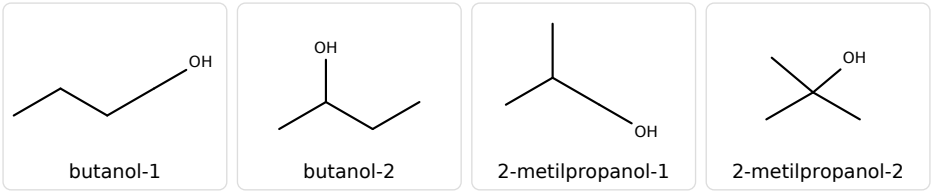


2.3 Izomeriyasi

Bir atomli spirtlarda uch xil izomeriya bor: **uglerod zanjiri**, **–OH ning o‘rni** va **sinflararo** (oddiy efirlar R–O–R’ bilan).

C_3H_8O – 3 ta izomer: 2 ta spirt (propanol-1, propanol-2) va 1 ta efir (metiletil efir CH_3–O–C_2H_5).

$C_4H_{10}O$ – 7 ta izomer: **4 ta spirt** va **3 ta efir** (dietil efir C_2H_5–O–C_2H_5, metilpropil efir, metilizopropil efir):



$C_5H_{11}OH$ tarkibli spirtlar – **8 ta**. Spirtlar izomerlari soni: C_1 – 1, C_2 – 1, C_3 – 2, C_4 – 4, C_5 – 8.

2.4 Fizik xossalari

- C_1–C_{11} spirtlar – **suyuqlik**, C_{12} va undan yuqorilari – qattiq moddalar.
- Metanol, etanol va propanollar suv bilan **istalgan nisbatda** aralashadi; uglerod soni ortgan sari eruvchanlik kamayadi.
- Molekulyar massa ortishi bilan qaynash harorati ortadi. **Vodorod bog‘ tufayli** spirtlar mos alkanlardan ancha yuqori haroratda qaynaydi.
- Izomerlarda tarmoqlangan zanjirli spirtning qaynash harorati past bo‘ladi.

Modda	Formula	M_r	t_{qayn}, °C
metanol	CH_3OH	32	64,7
etanol	C_2H_5OH	46	78,4
propanol-1	C_3H_7OH	60	97,2
propanol-2	$(CH_3)_2CHOH$	60	82,4
butanol-1	C_4H_9OH	74	117,7
etan (solishtirish)	C_2H_6	30	–88,6

Ehtiyot bo‘ling: metanol – kuchli zahar: 5–10 ml ichilsa ko‘rlik, 30 ml o‘limga olib keladi. Etanol narkotik ta’sir qiladi, tomir va jigarni zararlaydi; uni ichish zararli.

2.5 Olinishi

1) Alkenlarni gidratlash (Markovnikov qoidasi)

$$CH_2{=}CH_2 + H_2O \xrightarrow{H_3PO_4,\ 300\ °C,\ p} C_2H_5OH$$

$$CH_3{-}CH{=}CH_2 + H_2O \xrightarrow{H^+} CH_3{-}CH(OH){-}CH_3 \quad \text{propanol-2}$$